



Digital 3D Pain Mapping and Dermatome Analysis

Amit Bezalel¹, Shachar Guttman¹, Supervised by: Prof. Mickey Scheinowitz¹, Dr. Ifat Sandler²

¹ School of Biomedical Engineering, The Iby and Aladar Fleischman Faculty of Engineering, Tel Aviv University

² Pain Rehabilitation Unit, Loewenstein Rehabilitation Medical Center, Israel

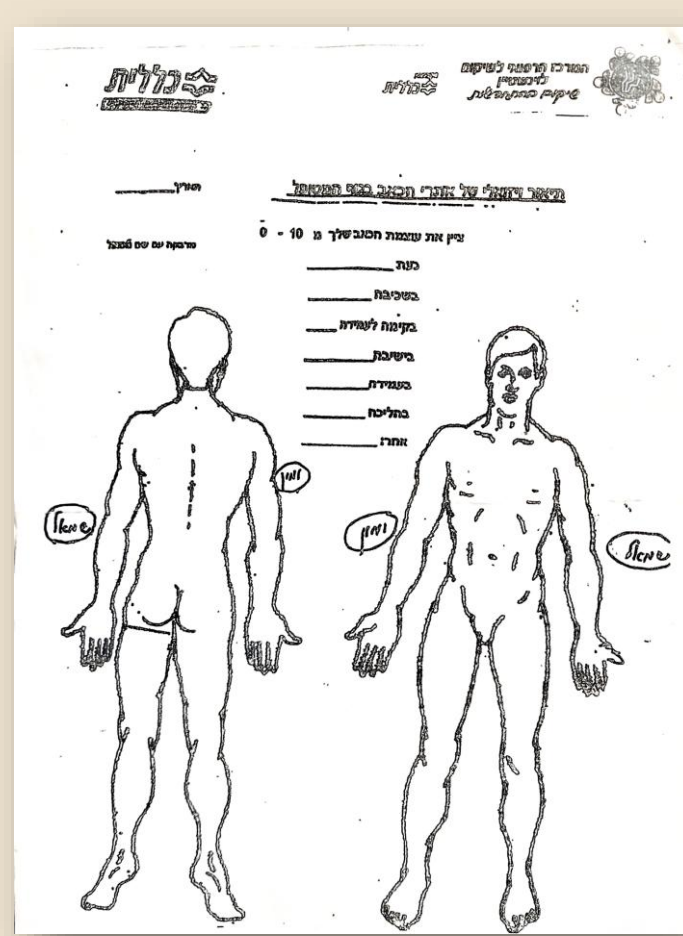
Introduction | The Clinical Problem

Pain is one of the most common reason people seek medical care. However, pain assessment still relies heavily on verbal descriptions, pain scales and paper body diagrams.

These methods limit spatial precision, structured documentation, and anatomical interpretation.

The way pain is represented can directly influence clinical interpretation;

accurate and comprehensive representation of pain improves communication, treatment planning and longitudinal follow-up.



Objectives



System development- Develop an interactive 3D model for accurate and comprehensive pain representation



Clinical decision support- Support clinical decision-making by integrating patient-reported pain with anatomical and clinical data



Clinical evaluation- Evaluate the system's feasibility, usability, and clinical potential in a proof-of-concept study

Methods

Digital System Development



Clinical Evaluation

Pain reporting methods:

- Traditional pen and paper drawing
- PAINDICATOR system

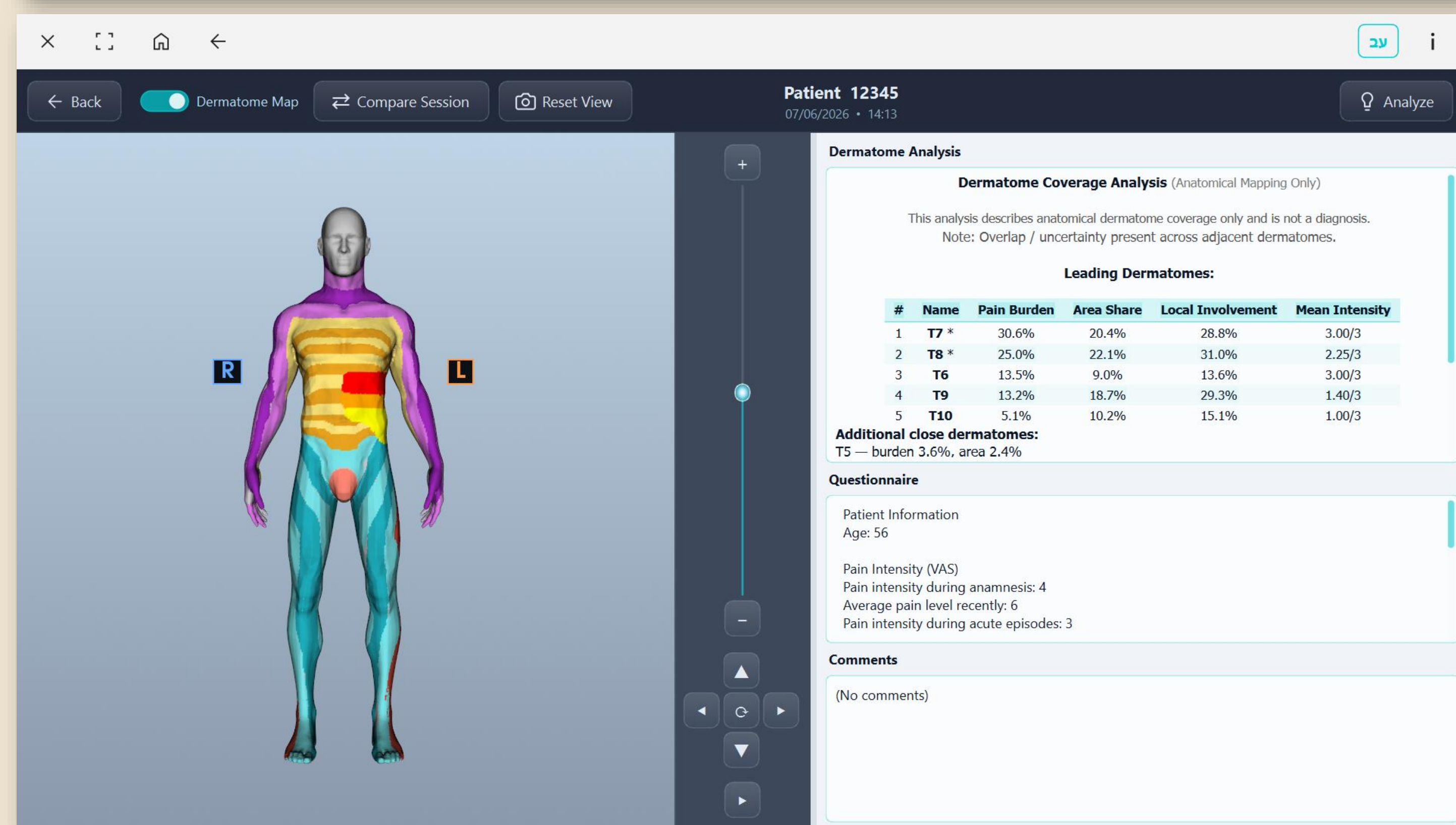
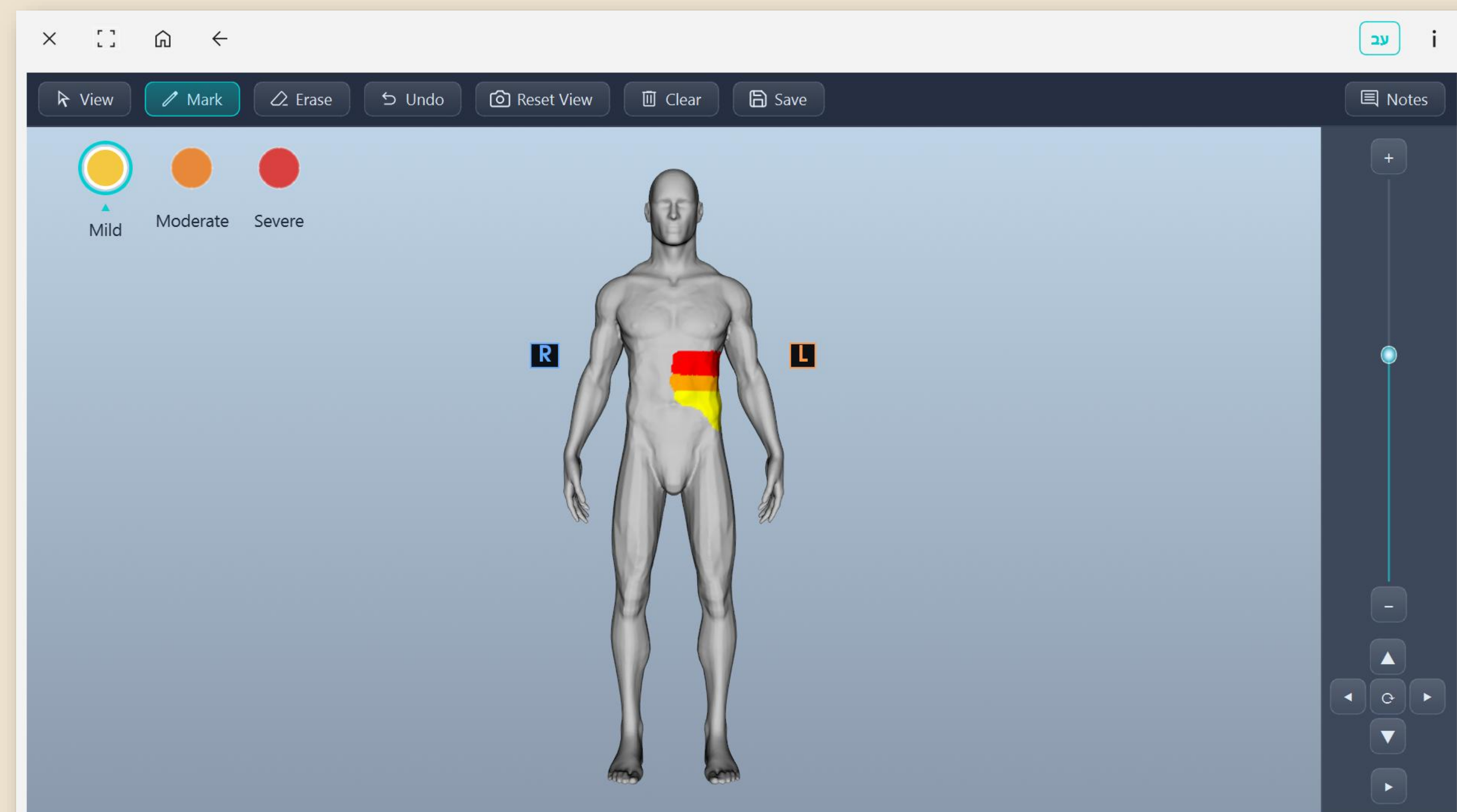
Collection and comparison:

- Usability and user experience measures
- Clinical interpretation and usefulness
- Completion time and workflow integration
- Overall satisfaction among users (patients and clinicians)



Results | System Capabilities

- ✓ Enables intuitive pain mapping on an **interactive 3D body model**
- ✓ Supports **pain location and pain intensity** documentation
- ✓ **Digitizes pain documentation**, enabling structured storage, longitudinal follow-up and clinical review
- ✓ **Integrates** questionnaire data and clinician notes
- ✓ Provides **dermatomal visualization** for informed interpretation
- ✓ **Analyzes data** and suggests the relevant dermatomes



Results | The Clinical Study

PAINDICATOR was evaluated at Loewenstein Rehabilitation Medical Center



PATIENTS (N=23)

Usability

The system is easy and clear to use

4.7/5

Pain Representation

Accurate representation of pain location and intensity

4.9/5

Future Use

Willingness to use the system with suitable patients

5/5

Overall Satisfaction

97%

4.9/5



CLINICIANS (N=23)

Usability

The system is easy and clear to use

4.2/5

Clinical Value

The system contributes to the clinical evaluation

4.5/5

Future Use

Willingness to use the system with suitable patients

5/5

Dermatome Evaluation

Dermatome assessment accuracy for relevant cases

3.8/5

Overall Satisfaction

94%

4.7/5

Summary and Conclusions

- **Digital transformation:** Replaces traditional paper pain drawings with structured digital documentation, enabling data storage, longitudinal tracking and improved clinical continuity.
- **Enhanced pain assessment:** Combines reported pain, questionnaire data and anatomical analysis to support the clinical evaluation.
- **Clinical validation:** High usability, feasibility and acceptance by patients and clinicians.